

In addition to the online homework, please hand in the following problems:

Problem 1: Suppose $\mathbf{F}$ conservative vector field and

$$g(x, y, z) = \int_{(1, -2, 0)}^{(x, y, z)} \mathbf{F} \cdot d\mathbf{r}$$

Show that $\nabla g = \mathbf{F}$

Problem 2: Which of the following vector fields are conservative.

- a. $\mathbf{F} = 2x\mathbf{i} + 3y\mathbf{j} + 4z\mathbf{k}$
- b. $\mathbf{F} = [y, x + z, -y]$
- c. $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$
- d. $\mathbf{F} = [e^{y+2z}, xe^{y+2z}, 2xe^{y+2z}]$

Problem 3: For the fields that were conservative in **Problems 2**, find the corresponding potential functions. (Show work)